



SOUND PLACEMENT

Building from Source - a Beginner's Guide

What you are about to do

Sound Placement is distributed as source code. To turn that source code into a working plugin, a **compiler** translates the code into a program, and **CMake** is the tool that organises the whole process. You do not need to understand C++ or CMake to build it - you only need to install two things and type two commands. This guide walks through everything, assuming no prior experience.

The build happens in two stages. First, the **configure** stage: CMake inspects your system, finds your compiler and downloads the JUCE framework (the library the plugin is built on) automatically - this is why the first run needs internet access and a few minutes. Second, the **build** stage: the compiler does its work and produces the plugin files.

Prerequisites

You need three things. Two are essential, one is optional:

1. A compiler

Windows	Install Visual Studio 2022 Community (free) from visualstudio.microsoft.com . During installation, tick the workload called " Desktop development with C++ " - this is the important part. The plain "Visual Studio Code" editor is a different product and is not enough.
macOS	Open the Terminal app and run <code>xcode-select --install</code> . Accept the dialog that appears - this installs Apple's command line tools.
Linux	Install the basics plus the libraries JUCE needs (Debian/Ubuntu shown): <code>sudo apt install build-essential cmake libasound2-dev libx11-dev libxext-dev libxrandr-dev libxinerama-dev libxcursor-dev libfontconfig1-dev</code>

2. CMake (version 3.22 or newer)

On **Windows**, Visual Studio 2022 already includes CMake, but the standalone installer from cmake.org/download is easier to use from a normal command prompt - during installation, choose "**Add CMake to the system PATH**". On **macOS**, install it with Homebrew: `brew install cmake` (get Homebrew from brew.sh if you do not have it). On **Linux**, the apt command above already installed it.

To check it works: open a terminal (on Windows: press the Windows key, type cmd, press Enter) and run:

```
cmake --version
```

If a version number appears, you are ready. If you get "command not found", CMake is not on your PATH - re-run the installer and pick the PATH option, then open a *new* terminal window.

3. Git (optional)

Git makes it easy to download the source and stay updated, but you can skip it: on the GitHub page, click the green **Code** button and choose **Download ZIP**, then unpack the ZIP somewhere convenient. If you do want Git, get it from **git-scm.com** (Windows/macOS) or your package manager (Linux).

Building, step by step

1 Get the source code

With Git (in a terminal):

```
git clone https://github.com/JanKar76/SoundPlacement.git
cd SoundPlacement
```

Without Git: download and unpack the ZIP as described above, then navigate your terminal into the unpacked folder. On Windows the easiest way: open the folder in File Explorer, click the address bar, type cmd and press Enter - a command prompt opens already in that folder.

2 Configure - let CMake set everything up

```
cmake -B build
```

This creates a folder called `build` containing all the generated project files, and downloads JUCE (roughly 400 MB) into it. Expect this to take several minutes the first time; later runs are fast because everything is cached. Nothing outside the `build` folder is touched.

Note: You only need to configure once. Re-run it if you delete the build folder or edit CMakeLists.txt.

3 Build - compile the plugin

```
cmake --build build --config Release
```

The compiler now works through JUCE and the plugin code. The first build takes 5-15 minutes depending on your machine; after that, only changed files are recompiled, so rebuilds take seconds. **Release** means an optimised build - this is what you want for actually using the plugin.

4 Find your files

When the build finishes you will find, inside the build folder:

VST3 plugin	build/SoundPlacement_artefacts/Release/VST3/Sound Placement.vst3
--------------------	--

Standalone app	build/SoundPlacement_artefacts/Release/Standalone/Sound Placement(.exe/.app)
-----------------------	--

The build also tries to copy the VST3 into your system plugin folder automatically (C:\Program Files\Common Files\VST3 on Windows, /Library/Audio/Plug-Ins/VST3 or ~/Library/Audio/Plug-Ins/VST3 on macOS, ~/.vst3 on Linux). Open your DAW, rescan plugins, and Sound Placement should appear. The standalone app is handy for a quick test without any DAW.

Troubleshooting

"cmake is not recognized" / "command not found"

CMake is not on your PATH. Re-run the CMake installer and select "Add CMake to the system PATH", or on Windows use the "x64 Native Tools Command Prompt for VS 2022" from the Start menu, which has everything preconfigured.

"No CMAKE_CXX_COMPILER could be found"

No C++ compiler is installed, or CMake cannot find it. On Windows, make sure the "Desktop development with C++" workload is installed in Visual Studio Installer. On macOS, run xcode-select --install.

Configure fails while downloading JUCE

The download needs internet access; check your connection, proxy or VPN. If it was interrupted halfway, delete the build folder and run the configure command again.

Copying to the VST3 folder fails (permission denied)

On Windows, writing to C:\Program Files may require an administrator terminal - or simply copy the .vst3 from the artefacts folder there yourself, once. Alternatively set COPY_PLUGIN_AFTER_BUILD to FALSE in CMakeLists.txt.

The build worked yesterday but fails today after changes

The safe reset is to delete the build folder entirely and redo steps 2-3. This costs a full rebuild but clears any stale cache.

Linux: errors about missing alsa/X11/freetype headers

One of the -dev packages from the prerequisites is missing. Re-run the full apt install command on page 1.

Glossary

CMake	A tool that generates build instructions for whatever compiler your platform uses. One project file works on Windows, macOS and Linux.
Configure	CMake's first stage: inspect the system, download dependencies, generate project files into the build folder.
Build	The compile stage: turn source code into the actual plugin binaries.
JUCE	The open source C++ framework the plugin is built on. Downloaded automatically during configure - you never install it yourself.

FetchContent	The CMake feature that performs that automatic download.
VST3	The plugin format most DAWs load. On disk it is a folder ending in .vst3.
Release / Debug	Two build flavours: Release is optimised for use, Debug is slower but easier to troubleshoot in a debugger.
Artefacts	CMake's word for the finished output files.

Questions or problems? Open an issue at github.com/JanKar76/SoundPlacement. Sound Placement is free software under the GNU AGPL v3.0.